



Analysis of immunological enhancement of immunosuppressed chickens by Chinese herbal extracts

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ARTICLE INFO

Article history:

Received 30 May 2009

Received in revised form 1 November 2009

Accepted 8 November 2009

Available online 13 November 2009

Keywords:

Chinese herbal extracts

Antibody titers

Immunosuppression

Reticuloendotheliosis virus

ABSTRACT

Ethnopharmacological relevance: *Radix astragali*, *Radix codonopsis*, *Herba epimedii* and *Radix glycyrrhizae* are 4 plants commonly used in Chinese traditional medicine or veterinary medicine to improve immune functions against chronic diseases in humans and animals.

Aim of the study: We compared immunological enhancement by 4 herbal extracts in clinical healthy chickens or immunosuppressed chickens singly and in combination.

Materials and methods: Water extracts of 4 herbs individually and in different combinations were supplemented in drinking water. Hemagglutination inhibition (HI) antibody titers against Newcastle disease virus (NDV) and H5 avian influenza virus (H5-AIV) after vaccination were measured as indicators to evaluate immunological stimulation across groups supplemented with different herbal extracts. The experiments were conducted in both clinically healthy chickens and chickens with immunosuppression induced by reticuloendotheliosis virus (REV) infection.

Results: In clinically healthy chickens HI antibody titers against NDV and H5-AIV after vaccination were not influenced by supplementation with the herbal extracts of *Radix astragali*, *Radix codonopsis*, *Herba epimedii* and *Radix glycyrrhizae* in drinking water. In chicks with REV-induced immunosuppression, however, supplementation of some herbal extracts significantly increased HI antibody titers to NDV and H5-AIV when compared to the immunosuppressed control group ($P < 0.01$), but the titers were still lower than those in chicks not infected by REV. The 4 herbal mixtures produced the best enhancement among various combinations. The components of the herbal extract were water soluble and treatment by ether had no influence on immunological enhancement. The molecular weights of the active components of the herbal extracts were in the range of 10,000–100,000 Da.

Conclusion: Our results show that the herbal extract supplementation in drinking water can induce an immune stimulation response in immunosuppressed chickens.

It suggests that chickens with REV infection-induced immunosuppression could be used as an experiment model for determination of immunological enhancement effects of some herbal components.

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1. Introduction

Many reports indicate that some Chinese herbal extracts can improve specific and nonspecific immunological reactions of mice, chicks, rabbits or pigs (Chu et al., 2004, 2005; Wang et al., 2004; Li et al., 2004; Hu et al., 2004; Wang et al., 2006). More reports were given to several herbs, such as *Radix astragali*, *Herba epimedii*, *Radix codonopsis* and *Radix glycyrrhizae* (Zhang and Wang, 1992, 1993; Xu et al., 2000; Liu et al., 2000; Cao et al., 2003; Ou et al., 2006; Shen et al., 2007; Teng et al., 2008; Shang et al., 2009; Luo et al., 2009). In experiments by Wang et al. (2006), supplement of some herb mixtures in the feeds improved the antibody responses in

chicks after vaccination against Newcastle Disease virus (NDV) or infectious Bursal disease virus (IBDV). Most experiments have been implemented on clinically healthy animals or immunosuppressed animals with administrated drugs such as corticosterone (Shen et al., 2007), florfenicol (Cao et al., 2003) or cyclophosphamide (Liu et al., 2000). In the poultry industry today, immunosuppression is common and generally induced by sub-clinical infection of viruses, such as reticuloendotheliosis virus (REV), Marek's disease virus (MDV), avian leukosis virus (ALV), chick infectious anemia virus (CAV) and avian reovirus (ARV). The REV infection often co-exists with other infections of ALV, MDV, CAV and ARV in chicken farms in China (Zhang et al., 2003; Cui et al., 2006, 2009), it makes immunosuppression more serious. There are demands in chicken industry for some herbs with immune enhancement effects. As measured by atrophy of Fabricius bursa and thymus gland, and HI antibody titers to NDV or AIV after immunization of inactivated

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vaccines, REV is one of the most powerful immunosuppressive viruses (Sun et al., 2006, 2009). Also, levels of immunosuppression induced by REV infection are controllable by doses and immunosuppressive state is stable. In this study, immunosuppression was induced by infecting chicks with a given dose of REV, to compare the immunological enhancement of selected Chinese herbal extracts.

2. Materials and methods

2.1. Composition of compound Chinese herbs

Radix astragali (*Astragalus membranaceus* Fisch, var. *mongholicus*, Leguminosae) as “A”, *Radix codonopsis* (*Codonopsis pilosula* Franch, Campanulaceae) as “C”, *Herba epimedii* (*Epimedium* spp., Berberidaceae) as “E” and *Radix Glycyrrhizae* (*Glycyrrhiza uralensis* Fisch, Leguminosae) as “G” were bought from Taian Medical Material Company, and identified by the Shandong Provincial Institute of Veterinary Drug Control.

We tested five combinations of the compounds. They were 40% A, 25% C, 25% E and 10% G (ACEG); 40% A, 30% C and 30% E (ACE); 50% A, 30% E and 20% G (AEG), 60% A and 40% C (AC) and extract A was purely *Radix astragali*.

2.2. Extract methods of water soluble and liposoluble substances from herbal mixtures

Both single and mixtures of herbs, were immersed overnight by using 8 times of water, then boiled for 2 h and filtered. Fivefolds of water were added into the sediments and boiled for 2 h, then filtered. The filtrates of two extracts were mixed, then boiled and concentrated, until a 1 ml solution contained 1 g of Chinese herb. To compare the activities of water soluble and liposoluble components of the Chinese herbs, herbal mixture ACEG was immersed in ether first and stirred twice daily, then the solution was filtered 1 week later. Glucose powder was added into the dry substance after ether was evaporated, 1 g mixed extract powder should contain ether-extracted components equivalent to 1 g Chinese herb. The mixtures were completely mixed and ground, passed through a 200-mesh sieve, and this extract was identified as ACEG-E, for the liposoluble components. The remaining herb mixture treated with ether was re-extracted with water as above and the water soluble extract was identified as ACEG-W.

2.3. Preparation of extracts in different ranges of molecular weights

The soluble ACEG extract was filtrated first by using an ultra-filtration membrane (pore size 100,000 Da, Tianjin Tonghai Water Purification Factory, TH-30-8). The components which did not pass through the filter membrane were labeled ACEG-L (molecular weight >100,000). Then the filtrate was filtrated again by using ultra-filtration membrane (pore size 10,000 Da, Tianjin Tonghai Water Purification Factory, TH-30-1); the components which did not pass through the filter membrane were identified as ACEG-M (10,000–100,000), and components which passed through the filter membrane were named as ACEG-S (<10,000). Each filtrate was then concentrated to the equivalent of the original 1 g herbal mixture per ml.

2.4. Animals and treatments

One-day-old males of Hy-line brown breed were used in present study. To evaluate the maternal antibodies to REV, serum samples were collected from 40 chicks. REV antibody was detected by using REV antibodies detection kit (IDEXX Laboratory, USA), and

the positive rate of antibody to REV was 2.3%. One-day-old specific pathogen free (SPF) chickens were bought from Jinan SPAFS Company. To maintain SPF conditions, the chicks were kept in isolators with filtered air of positive pressure. Drinking water and feeds were autoclaved before use.

REV-C99-p3 is a molecularly cloned strain preserved in our laboratory (Ji et al., 2005). Diluted REV solution (500TCID₅₀ for each chick) was injected into the muscle of inner thigh in chicks at 1 day of age.

Chicks at 7 days of age were vaccinated with live NDV vaccine of Lasota strain by eye-drop, and inactivated vaccines of NDV (Intervet) and H5-AIV (Qilu Animal Health Products Co., Ltd.) were injected intramuscularly. The H5-AIV vaccine contained concentrated antigen. HI antibody levels to NDV and H5-AIV were determined 6 weeks after immunization.

2.5. Design of animal experiment

Five independent experiments were conducted to assess the different factors affecting immunological enhancement of herbal extracts.

In experiment 1, we compared the immunological enhancement effect of Chinese herbal extracts on normal and immunosuppressed chicks. There were 120 1-day-old Hy-line brown male chicks divided randomly into 4 groups. Groups 1 and 2 were kept in normal conditions and not infected with REV; groups 3 and 4 were infected with 500 TCID₅₀ REV to induce immunosuppression. From 1 day until the end of the experiment, ACEG extract was added to the drinking water at the rate of 2% for groups 2 and 4, but not for groups 1 and 3 as controls.

Experiment 2 was for comparisons of effects of herbal extracts under different conditions in commercial regular chickens and SPF-origin chickens. 180 1-day-old SPF chickens were randomly divided into 6 groups. Chickens in groups 1–3 were maintained in isolators with filtered air; drinking water and feeds were autoclaved. Chicks of SPF-origin in groups 4–6 were kept in regular non-SPF clean room with regular water and feeds. Another 90 Hy-line brown chicks were divided into groups 7–9. Group 1, 4 and 7 were used as controls without immunosuppression; groups 2, 3, 5, 6, 8 and 9 were infected with REV to induce immunosuppression. For groups 3, 6 and 9, the herbal extract ACEG was added into drinking water at the rate of 2% from the 1st day until the end of experiment. The other groups were not fed with ACEG as controls.

Experiment 3, compared herbal extracts of different combinations for their immunological enhancement. 210 Hy-line brown chickens were divided into 7 groups. Group 1 was the control group with no REV infection, and groups 2–7 were immunosuppressed chickens by REV infection. No herbal extract was added to the drinking water in groups 1 and 2, but different herbal extract combinations ACEG, ACE, AEG, AC and A were added to the drinking water at the rate of 2% during the whole period for groups 3, 4, 5, 6 and 7.

In experiment 4, we compared the immunological enhancement between water soluble and liposoluble extracts. 150 Hy-line brown chicks were divided into 5 groups. Groups 2–5 were immunosuppressed chickens by REV infection, and group 1 was clinically healthy chickens with no REV infection. Groups 1 and 2 were controls with no herbal extract added to drinking water. During the whole period of the experiment, ACEG and ACEG-W were added into the drinking water at 2% for groups 3 and 4, ACEG-E powder was mixed into the feeds at 4% for group 5.

Experiment 5 was used to determine the molecular weight range of the active components in the herbal extracts. 180 Hy-line brown chickens were divided into 6 groups. Groups 2–6 were immunosuppressed chickens by REV infection; group 1 was

Table 1

Influence of herbal extract ACEG on immunological responses in REV infected and control chicks (Exp. 1).

Groups	REV infected	Herbal extract	HI antibody level 6 weeks after immunization ($\log^2$)	
			To NDV	To H5-AIV
1	–	/	6.35 $\pm$ 0.92 ^a (n = 26)	6.88 $\pm$ 0.97 ^a (n = 26)
2	–	ACEG	6.38 $\pm$ 1.00 ^a (n = 26)	6.46 $\pm$ 0.89 ^a (n = 26)
3	+	/	3.21 $\pm$ 1.00 ^c (n = 24)	3.25 $\pm$ 0.97 ^c (n = 24)
4	+	ACEG	4.36 $\pm$ 1.09 ^b (n = 25)	4.36 $\pm$ 0.84 ^b (n = 25)

Note: The values in the table were means $\pm$ SD (n = sample size). The data in the same columns were compared, different small letters at the upper-right corner indicates significant difference ($P < 0.05$).

clinically healthy chickens without REV infection. ACEG, ACEG-L, ACEG-M and ACEG-S representing components in different molecular weight ranges were added to the drinking water at 2% for groups 3–6, and groups 1 and 2 were controls without any herbal extract added to the drinking water.

2.6. Determination of REV viremia levels in chick blood

To investigate whether the herbal extract influenced the replication of REV in chicks, 12 1-day-old SPF chicks were divided into 2 groups and kept in two isolators with ordinary feeds and water. For chickens in group 1, 2% ACEG was added in drinking water, and the other group was the control group. After infection with REV of 500 TCID₅₀, viremia levels were determined in each chick at ages of 10, 20 and 30 days respectively, following previously reported protocols (Zhuang et al., 2006).

2.7. Statistical analysis

All data were expressed in $\bar{X} \pm SD$, and the difference between groups was analyzed by using Student's *t*-test (Excel).

3. Results

3.1. Comparison of immunological enhancement of herbal extract ACEG on immunosuppressed and control chicks

Table 1 indicates that, herbal extract ACEG added to drinking water in clinically healthy chickens had no effect on antibody response ($P > 0.05$), for HI antibody titers of NDV or H5-AIV (groups 1 and 2 in Table 1 6 week after vaccination with inactivated vaccination at the age of 7 days). However, supplement of herbal extract ACEG improved immunological reaction in immunosuppressed chickens, HI antibody titers to NDV and H5-AIV were significantly higher in group 4 with extract ACEG than that in group 3 without ACEG supplement ($P < 0.05$), although still lower than the control group 1.

3.2. Comparison of immunological enhancement of herbal extracts ACEG in SPF chicks and regular commercial chicks

As Table 2 demonstrated, the herbal extract ACEG did not increase antibody reactions in immunosuppressed chicks of SPF-origin kept under strict SPF conditions (between groups 2 and 3). However, ACEG increased HI antibody titers significantly ($P < 0.05$) to NDV and H5-AIV in similar REV infected immunosuppressed chicks of SPF-origin if kept under regular conditions (between groups 5 and 6) as non-SPF commercial regular chicks (between groups 8 and 9, $P < 0.01$). But the antibody titers in chicks with ACEG were still significantly lower than their normal control chicks without REV-induced immunosuppression (between groups 4 and 6, or groups 7 and 9). Results of experiment 2 indicated that supplement of herbal extract ACEG increased antibody reactions in REV infected immunosuppressed chicks kept under regular conditions but not in chicks of SPF-origin kept under SPF condition with autoclaved water and feeds and filtered air. It suggests that the immunological enhancement effect of ACEG supplemented in water may depend on the microbial status inside chick intestines.

3.3. Comparisons of different herbal mixtures for their immunological enhancement on chicks

The results of experiment 3 demonstrated that immunosuppression was induced in 1-day chicks infected with REV. As Table 3 shows there was significant difference between the control and REV-infected chicks (groups 1 and 2, $P < 0.01$). In REV-induced immunosuppressed chicks, 5 different herbal extract mixtures were supplemented in drinking water respectively (groups 3–7). Analysis of HI antibody titers to NDV and H5-AIV (Table 3) demonstrated immunological enhancement effects by two herbs (AC, group 6), three herbs (ACE and AEG, groups 4 and 5), or 4 herbs (ACEG, group 3) but not for group 7 with extract from one herb A. In the response to NDV, HI antibody titers in chick groups 2–6 were all significantly higher than control group 2 without herbal extract supplement and group 7 supplemented with the extract from only A ($P < 0.05$). In antibody responses to H5-AIV, only herbal extracts ACEG (group 3) and ACE (group 4) gave significantly higher HI titers

Table 2

Comparisons of effects of herbal extracts on antibody responses under different conditions in SPF-origin chickens and regular commercial chickens (Exp. 2).

Groups	REV infection	Herbal extract	Sources of chicks	If under SPF condition	HI antibody level 6 weeks after immunization ($\log^2$)	
					To NDV	To H5-AIV
1	–	/	SPF-origin	Yes	6.74 $\pm$ 0.93 ^a (n = 27)	6.93 $\pm$ 0.81 ^a (n = 27)
2	+	/	SPF-origin	Yes	3.13 $\pm$ 0.80 ^c (n = 24)	3.04 $\pm$ 0.79 ^c (n = 24)
3	+	ACEG	SPF-origin	Yes	3.28 $\pm$ 0.29 ^c (n = 25)	2.92 $\pm$ 0.98 ^c (n = 25)
4	–	/	SPF-origin	No	6.12 $\pm$ 0.90 ^a (n = 28)	6.46 $\pm$ 0.98 ^a (n = 28)
5	+	/	SPF-origin	No	3.00 $\pm$ 0.75 ^c (n = 25)	2.80 $\pm$ 0.69 ^c (n = 25)
6	+	ACEG	SPF-origin	No	4.04 $\pm$ 0.79 ^b (n = 27)	4.04 $\pm$ 0.92 ^b (n = 27)
7	–	/	Hy-line	No	6.62 $\pm$ 0.79 ^a (n = 26)	6.54 $\pm$ 0.63 ^a (n = 26)
8	+	/	Hy-line	No	3.04 $\pm$ 0.89 ^c (n = 24)	2.92 $\pm$ 0.76 ^c (n = 24)
9	+	ACEG	Hy-line	No	4.36 $\pm$ 0.93 ^b (n = 25)	3.72 $\pm$ 0.87 ^b (n = 25)

Note: As in Table 1.

Table 3

Effects of different herbal extract mixtures on HI antibody titers to NDV and H5-AIV in chicks immunosuppressed by REV infection (Exp. 3).

Groups	REV infection	Herbal extract	HI antibody level 6 week after immunization ($\log^2$)	
			To NDV	To H5-AIV
1	–	/	6.54 ± 0.87^a ($n=28$)	7.18 ± 0.89^a ($n=28$)
2	+	/	3.14 ± 0.76^c ($n=23$)	3.83 ± 0.82^c ($n=23$)
3	+	ACEG	4.36 ± 0.79^b ($n=25$)	4.68 ± 0.79^b ($n=25$)
4	+	ACE	4.17 ± 1.11^b ($n=24$)	4.67 ± 0.90^b ($n=24$)
5	+	AEG	4.13 ± 1.10^b ($n=24$)	4.04 ± 0.93^c ($n=24$)
6	+	AC	4.28 ± 1.04^b ($n=25$)	4.08 ± 0.80^c ($n=25$)
7	+	A	3.20 ± 0.93^c ($n=25$)	3.76 ± 0.89^c ($n=25$)

Note: As in Table 1.

Table 4

Effects of water soluble and liposoluble components on HI antibody titers to NDV and H5-AIV in chicks immunosuppressed by REV infection (Exp. 4).

Groups	REV infection	Herbal extract supplemented with	HI antibody titers 6 weeks after immunization ($\log^2$)	
			To NDV	To H5-AIV
1	–	/	5.93 ± 1.15^a ($n=27$)	5.59 ± 1.10^a ($n=27$)
2	+	/	1.91 ± 0.83^c ($n=23$)	2.13 ± 0.85^c ($n=23$)
3	+	ACEG	3.32 ± 1.16^b ($n=25$)	3.64 ± 0.79^b ($n=25$)
4	+	ACEG-E	2.36 ± 1.09^c ($n=25$)	2.64 ± 0.89^c ($n=25$)
5	+	ACEG-W	3.31 ± 1.46^b ($n=26$)	3.15 ± 1.23^b ($n=26$)

Note: As in Table 1.

than control group 2 ($P < 0.05$). HI titers in chicks of groups 5 (AEG) and 6 (AC) were higher than control group 2 but the differences were not significant ($P > 0.05$).

3.4. Comparison of immunological enhancement of water soluble and liposoluble components

Immunological enhancement effects of water soluble or liposoluble components of herbal extracts were also compared in chicks immunosuppressed by REV infection. As Table 4 indicates, water soluble extract (ACEG-W) from herb ACEG treated by ether maintained its immunological enhancement effect in immunosuppressed chickens like the original extract ACEG. HI titers to NDV and H5-AIV in chicks supplemented with ACEG (group 3) or ACEG-W (group 5) were significantly higher than control group 2 ($P < 0.05$). There was no significant difference between groups 3 and 5. However, HI titers in group 4 supplemented in feeds with ether extract of ACEG was significantly lower than groups 3 and 5 supplemented with water soluble extracts. There was no significant difference in HI titers between group 4 and control group 2 ($P > 0.05$). It suggests the immunological enhancement of ACEG may be mainly due to water soluble components.

3.5. Comparison of immunological enhancement of active components with different molecular weight ranges

Table 5 shows the difference in immunological enhancement of components with different molecular weights in the herbal extract

Table 5

Immunological enhancement effects of herbal extract ACEG within different molecular weights on HI antibody titers to NDV and H5-AIV in chicks immunosuppressed by REV infection (EXP.5).

Groups	REV infection	Herbal extract	HI antibody titers 6 weeks after immunization ($\log^2$)	
			To NDV	To H5-AIV
1	–	/	6.00 ± 0.78^a ($n=26$)	6.04 ± 1.04^a ($n=26$)
2	+	/	2.92 ± 0.81^c ($n=24$)	3.04 ± 0.98^c ($n=24$)
3	+	ACEG	3.62 ± 0.96^b ($n=26$)	4.38 ± 0.92^b ($n=26$)
4	+	ACEG-L	2.84 ± 0.88^c ($n=25$)	3.24 ± 1.03^c ($n=25$)
5	+	ACEG-M	3.80 ± 0.89^b ($n=25$)	4.44 ± 0.98^b ($n=25$)
6	+	ACEG-S	2.84 ± 0.88^c ($n=25$)	2.96 ± 0.96^c ($n=25$)

Note: As in Table 1.

ACEG when studied in chicks immunosuppressed by REV infection. As the original ACEG (group 3), the filtrated ACEG-M (filtered through the ultra-filtration membrane with pore size 100,000 Da but not with 10,000) significantly increased HI antibody levels to NDV and H5-AIV (groups 5) when compared to the control group 2 ($P < 0.01$). But ACEG-L with larger molecular weights (group 4) and ACEG-S with smaller molecular weights (group 6) had no immunological enhancement (compared to group 2, $P > 0.05$). The results suggest that the active components with immunological enhancement of ACEG are located in the molecular weight range between 10,000 and 100,000 Da.

3.6. Effects of herbal extracts on viral replication

After 1-day chicks were infected with REV, the viremia levels in 6 chicks supplemented with herbal extract ACEG and 6 control chicks were measured. There was no significant difference in REV viremia levels 10, 20 and 30 days after infection (data omitted).

4. Discussion

Wang et al. (2006) has reported that herbal extracts can improve immunological response in chicks. However, the herbal extracts used in this study did not show immunological enhancement response in normal chicks. Rather, we found that compound mixtures enhanced immunological responses in immunosuppressed chicks. It suggests that some herbal components may have their immune enhancement effects only in immunosuppressed animals.

Actually, there were some reports indicating immune enhancement of *Radix astragali* and *Herba epimedii* in chickens and rats with immunosuppression induced by some chemicals or drugs such as corticosterone, florfenicol or cyclophosphamidum (Liu et al., 2000; Cao et al., 2003; Shen et al., 2007). In this study, an immunosuppressed chicken model induced by REV infection was used to study immune enhancement of some herb extracts. It is more similar to natural conditions, since the immunosuppression induced by REV infection is very common in chicken farms in China (Cui et al., 2006, 2009; Sun et al., 2009). REV is one of most strong immunosuppressive viruses in chickens, but mechanism for REV to induce immunosuppression is not very clear. It was recognized that REV infection causes atrophy of lymphocyte follicles in central immune organs such as thymuses and the Bursa. But it is not clear if ACEG can prevent thymuses and the Bursa from atrophy in REV-induced immunosuppressed chickens. Further study in the aspect should be done.

The results demonstrated that herbal extract from all 4 herbs or ACE extracted from *Radix astragali*, *Radix codonopsis* and *Herba epimedii* had immunological enhancement activity (Table 3). In responses to H5-AIV, immunological enhancement effects of AEG or AC were significantly lower than ACEG and ACE (Table 3). It is noted that herbal extract A (*Radix astragali*) alone had no immunological enhancement, even when the concentration of *Radix astragali* was higher or lower than that in the mixtures of ACEG or ACE (data omitted). This is obviously inconsistent with the reported immunological enhancement effect of astragalan extract for injection. It is not clear whether this is due to different administration method. However, comparisons in the study also demonstrated some synergism when different herb extracts were used in combination in REV-induced immunosuppressed chickens as in other animal experiments (Cao et al., 2003; Shen et al., 2007).

Usually, experiments with SPF-origin chickens under strict SPF conditions give constant and reliable results because it can prevent other inference from the environment. Contrary to our hypothesis, the same herbal extracts had no effects in SPF-origin chicks when kept under a strict SPF conditions with SPF feedings (Table 2). However, immunological enhancement of the herbal extract ACEG was found in the similar SPF-origin chicks immunosuppressed by REV infection when kept under regular conditions with regular feedings and drinking water (Table 2). Our results indicate that the use of filtered air and sterilized drinking water and feeds under strict SPF conditions dramatically changed the microbial population in the chicken digestive tracts. Different microbes in the intestines may produce different enzymes, which may cause different enzymatic and biochemical reactions to original herbal components taken into intestines. Thus, the same chemical compounds from herbs may change in different ways when chickens are kept under regular condition or strict SPF condition with filtered air, sterilized water and feeds. Different derivatives from the same original herbal compounds may demonstrate different effects on chicken immune responses when absorbed into the blood. Although the mechanism of interaction between herbal extracts and intestinal microbial biota is not clear, it is recommended that SPF-origin chickens under strict SPF conditions is not fit for studies of orally administrated herbs for their pharmaceutical activity.

This study confirms that the immunological enhancement of ACEG is due to water soluble components with molecular weight between 10,000 and 100,000 Da (Tables 4 and 5), it could help to further identify effective molecules. Actually, some chemical components from these 4 herbs were demonstrating their immunological enhancement effects, such as total flavones of epinedium from *Herba epimedii*, Epinedium polysaccharides; glyceric acid from *Radix glycyrrhizae*, polysaccharides from *Radix astragali* and *Radix codonopsis*, especially in combinations (Zhang and Wang, 1993; Xu et al., 2000; Shen et al., 2007; Shang et al.,

2009). We are not sure if these compounds or any other molecules were responsible for the immune enhancement in this animal experiment model, further studies are needed. No matter the extract ACEG was prepared by extraction of the mixture of four herbs A, C, E and G first, or extracts were made from each of 4 herbs separately and then mixed 4 extracts from single herbs into the extract of ACEG, the immune enhancement effects were exactly same (data omitted). It suggests that interaction between different components in ACEG did not occur during extraction processes, but the synergistic effects on immune enhancement occurred in vivo of chicks.

It would be better if the immune enhancement effects of ACEG and other combinations were also compared to some known immune stimulators as positive controls in REV-induced immunosuppressed chickens. Unfortunately, there is no such licensed or registered chemical or drug available in the market for chickens. Although levamisole was widely recognized and studied as a potential immune stimulant compound in many pharmacology text books and literatures (Ishikawa and Shimizu, 1983; Giambrone and Klesius, 1985; Božić et al., 2006; Janjatović et al., 2008; Pekmezci and Cakiroglu, 2009), it did not demonstrate any immune enhancement effect in our study, no matter in SPF chickens or REV-induced immunosuppressed chickens (data are not included). Actually, there were some controversial reports on its immune enhancement (Van Der Maaten et al., 1983; Roth and Kaeberle, 1984; Purswell et al., 1998).

In conclusion, we demonstrated that supplementation of *Radix astragali*, *Radix codonopsis*, *Herba epimedii* and *Radix glycyrrhizae* extracts in drinking water improves immune response and increase antibody titers to NDV and H5-AIV after vaccination in immunosuppressed chickens. But the same herbal extract did not show such immunological enhancement in clinically healthy chickens. The results indicate that different herbal extracts may have different mechanisms in immunological enhancement. Therefore, different animal models are needed to test the immunological enhancement for different herbal extracts. Our study suggests that chickens immunosuppressed by REV infection could be used as a animal experimental models for testing immunological enhancement for some herbs and their fractional components.

Conflict of interest

The authors declare that there are no conflicts of interest.

Acknowledgments

The study was supported by National Natural Science Foundation of China, fund #30330450. Authors thank Dr. David Buchanan, University of Massachusetts Amherst, USA, for English language improvement.

References

- Božić, F., Lacković, G., Kovsca-Janjatović, A., Smolec, O., Valpotić, I., 2006. Levamisole synergizes experimental F4ac+ *Escherichia coli* oral vaccine in stimulating ileal Peyer's patch T cells in weaned pigs. *Journal of Veterinary and Pharmacology Therapy* 29, 199–204.
- Cao, J.Y., Zhou, H.B., Lu, X.C., Dou, S.L., Liu, Y.W., Li, X.B., 2003. The effects of florfenicol and Chinese herbal ingredients *Radix astragali* and *Herba epimedii* on immune response in chicks. *Acta Veterinaria et Zootechnica Sinica* 34, 412–416.
- Chu, Y.F., Li, X.R., Hu, Y.L., 2004. Effects of Chinese herbal medicinal ingredient on cells mediated immunity in mice. *Journal of Nanjing Agricultural University* 27, 100–107.
- Chu, Y.F., Yan, X.M., Hu, Y.L., 2005. Immuno-enhancing effects of nine Chinese herbal medicine ingredients. *Chinese Journal of Veterinary Science and Technology* 35, 67–70.
- Cui, Z.Z., Meng, S.S., Jiang, S.J., Wei, J.P., 2006. Serological surveys of chicken anemia virus, avian reticuloendotheliosis virus and avian reovirus infections in white meat-type chickens in China. *Acta Veterinaria et Zootechnica Sinica* 37, 152–157.

- Cui, Z.Z., Sun, S.H., Zhang, Z., Meng, S.S., 2009. Simultaneous endemic infections with subgroup J avian leukosis virus and reticuloendotheliosis virus in commercial and local breeds of chickens. *Avian Pathology* 38 (6), 443–448.
- Giambone, J.J., Klesius, P.H., 1985. Effect of levamisole on the response of broilers to coccidiosis vaccination. *Poultry Science* 64, 1083–1089.
- Janjatović, A.K., Lacković, G., Božić, F., Popović, M., Valpotić, I., 2008. Levamisole synergizes proliferation of intestinal IgA+ cells in weaned pigs immunized with vaccine candidate F4ac+ nonenterotoxigenic *Escherichia coli* strain. *Journal of Veterinary and Pharmacology Therapy* 31, 328–333.
- Hu, Y.L., Kong, X.F., Li, X.R., 2004. Effects of ten Chinese herbal medicinal ingredients on proliferation and resisting NDV infection of CEF. *Acta Veterinaria et Zootechnica Sinica* 35, 301–305.
- Ishikawa, H., Shimizu, T., 1983. Depression of B-lymphocytes by mastitis and treatment with levamisole. *Journal of Dairy Science* 66, 556–561.
- Ji, R., Cui, Z.Z., Wang, X.L., 2005. Study of the infectivity of the molecular cloned reticuloendotheliosis virus and its genome. *Chinese Journal of Virology* 21, 448–455.
- Li, Y.H., Tian, G.B., Li, Y.B., 2004. Mechanism of immunity on anti-Avian influenza virus of complex Chinese traditional medicine. *Chinese Journal of Preventive Veterinary Medicine* 26, 416–419.
- Liu, J.G., Hu, Y.L., Zhang, B.K., Song, D.L., 2000. Immunomodulating action of epimedium-propolis adjuvant. *Chinese Journal of Veterinary Science* 20, 383–385.
- Luo, Y., Shao, Y.B., Gu, X.L., Li, X.Y., 2009. Effects of epimedium herb polysaccharides on the immunity function and vaccination in chickens. *China Poultry* 31, 22–26.
- Ou, Y.S.Z., Wang, S.S., Tian, S.X., Liang, X.L., Cai, M., 2006. Study of epimedium propolis mixture on prompting cell immunity of chicken. *Acta Veterinaria et Zootechnica Sinica* 37, 80–83.
- Pekmezci, D., Cakiroglu, D., 2009. Investigation of immunomodulatory effects of levamisole and vitamin E on Immunity and some blood parameters in newborn Jersey calves. *Veterinary Research and Communication* 33, 711–721.
- Purcell, B.J., Dawe, D.L., Brown, J., Williams, D.J., 1998. Effect of levamisole on immune function and reproductive performance in first-litter gilts. *American Journal of Veterinary Research* 49, 856–859.
- Roth, J.A., Kaeblerle, M.L., 1984. Effect of levamisole on lymphocyte blastogenesis and neutrophil function in dexamethasone-treated cattle. *American Journal of Veterinary Research* 45, 1781–1784.
- Shang, C.C., Zhao, Y., Mu, Y., Liang, N., Zhou, L.M., Qiu, M.N., Wang, X.K., Li, N., 2009. Effects of epimedium polysaccharide on immune organs and immune effects of Newcastle disease vaccine 34. *Progress in Veterinary Medicine* 34, 200–203.
- Shen, Z.Y., Wu, Z.J., Yu, L.H., Fu, D.X., Huang, J.H., 2007. Study the protection and combination on the total flavones of epimedium to the immunodeficiency rats induced by corticosterone. *Chinese Pharmacological Bulletin* 23, 1061–1063.
- Sun, S.H., Cui, Z.Z., Qiu, Y.Y., 2006. Immuno-suppression of reticuloendotheliosis virus infection on antibody responses to different vaccines in SPF chicks. *Virologica Sinica* 21, 34–37.
- Sun, S.H., Cui, Z.Z., Wang, J., Wang, Z.L., 2009. Protective efficacy of vaccination against highly pathogenic avian influenza is dramatically suppressed by early infection of chickens with *reticuloendotheliosis*. *Avian Pathology* 38, 31–34.
- Teng, F., Zeng, Y.Y., Huang, X.M., Yang, Z., Yao, M.L., Song, B., Li, L., 2008. Effect of extract from *Herba emidemiion* activation and proliferation of mouse T lymphocytes in vitro. *Chinese Traditional and Herbal Drugs* 39, 1355–1359.
- Van Der Maaten, M.J., Schmerr, M.J.F., Miller, J.M., Sacks, J.M., 1983. Levamisole does not affect the virological and serological responses of bovine leukemia virus-infected cattle and sheep. *Canadian Journal of Comparative Medicine* 47, 474–479.
- Wang, D.Y., Hu, Y.L., Kong, X.F., 2004. Effects of Chinese herbal medicinal ingredients on peripheral T lymphocyte transformation in chicks. *China Journal Veterinary Science* 4, 578–580.
- Wang, F.C., Fang, C.G., Zhang, Y.H., 2006. Effects of compound Chinese herbal immune synergist on immunity of egg-laying chicks. *Acta Veterinaria et Zootechnica Sinica* 37, 187–192.
- Xu, Y., Mu, X.S., Wang, G., Sun, P.Y., Zhang, B.F., Chen, Y.J., 2000. The effects of epimedium polysaccharides on the immune function of immune-depressed mice. *Journal of Shenyang Pharmaceutical University* 17, 434–437.
- Zhang, R.X., Wang, F.L., 1993. Effects of CPPS on humoral immunity and IL-2 production. *Journal of Lanzhou Medical College* 19, 14–17.
- Zhang, R.X., Wang, F.L., 1992. Regulatory effects of CPPS on cell-mediated immunity in mice. *Journal of Lanzhou Medical College* 18, 161–165.
- Zhang, Z., Cui, Z.Z., Jiang, S.J., 2003. Dual-infection of Marek's disease virus and reticuloendotheliosis virus from tumors in chickens. *Chinese Journal of Preventive Veterinary Medicine* 25, 275–278.
- Zhuang, G.Q., Sun, S.H., Cui, Z.Z., 2006. Interaction of Marek's disease virus and reticuloendotheliosis virus in co-infected broilers. *Virologica Sinica* 21, 157–162.